4. VACUUM CLEANER PROBLEM

Aim:

To develop a Python program that simulates the Vacuum Cleaner Problem using a simple reflex agent to clean all dirty rooms.

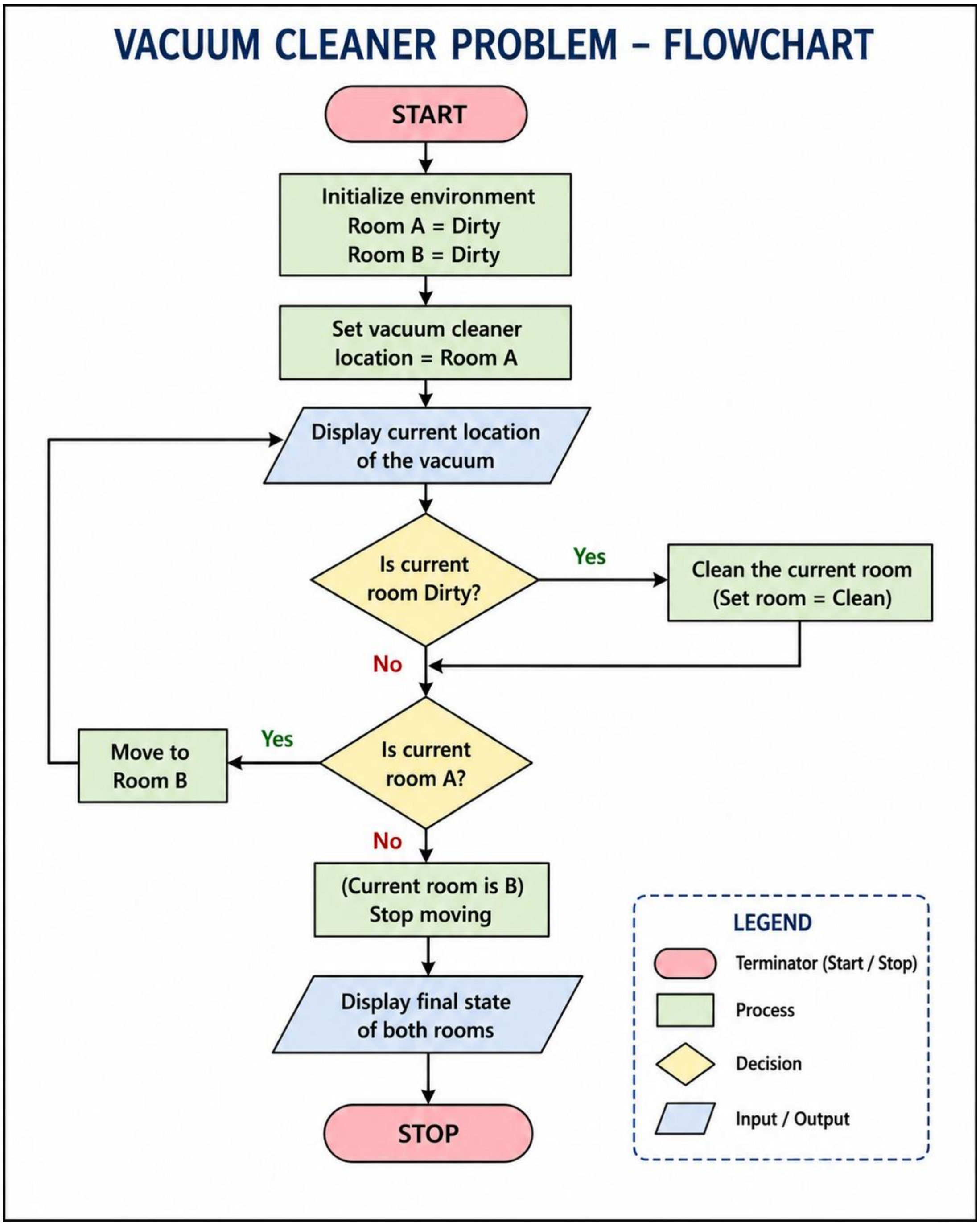
Objective :

- To understand the working of a simple reflex agent.
- To simulate the behavior of a vacuum cleaner in an environment.
- To clean dirty rooms based on their current state.
- To display the actions performed by the vacuum cleaner.

Algorithm:

1. Start.
2. Define the environment with two rooms (A and B).
3. Set the initial location of the vacuum cleaner.
4. Check the current room.
5. If the room is dirty:
   - Clean the room.
6. Otherwise:
   - Move to the next room.
7. Repeat until both rooms are clean.
8. Display the final state of the rooms.
9. Stop.

Flowchart :



Code:

```
rooms = {
    "A": "Dirty",
```

```
    "B": "Dirty"
}
location = "A"
while True:
    print("\nVacuum is in Room", location)
    if rooms[location] == "Dirty":
        print("Cleaning Room", location)
        rooms[location] = "Clean"
    if location == "A":
        location = "B"
    else:
        break
print("\nFinal State")
for room in rooms:
    print("Room", room, ":", rooms[room])
print("\nAll rooms are clean.")
```

Output:

```
Output

Vacuum is in Room A
Cleaning Room A

Vacuum is in Room B
Cleaning Room B

Final State
Room A : Clean
Room B : Clean

All rooms are clean.

=== Code Execution Successful ===
```

Result :

The Vacuum Cleaner Problem was successfully implemented using Python. The vacuum cleaner visited each room, cleaned the dirty rooms, and finally ensured that all rooms were clean.